

2	(a) ball moving in <u>opposite</u> direction (after collision)	B1	[1]
	(b) (i) change in momentum = $1.2 (4.0 + 0.8)$	C2	
	(correct values, 1 mark; correct sign {values added}, 1 mark)		
	= 5.76 N s ...(allow 5.8)	A1	[3]
	(ii) force = $\Delta p / \Delta t$ or $m\Delta v / \Delta t$	C1	
	= $5.76 / 0.08$ or $1.2 \times 4.8 / 0.08$	C1	
	= 72 N	A1	[3]
	(c) $5.76 = 3.6 \times V$	C1	
	$V = 1.6 \text{ m s}^{-1}$	A1	[2]
	(d) either speed of approach = 4.0 m s^{-1} and		
	speed of separation = 2.4 m s^{-1}	M1	
	not equal and so inelastic	A1	
	or kinetic energy before = 9.6 J and		
	kinetic energy after collision = 4.99 J	M1	
	kinetic energy after is less / not conserved so inelastic	A1	[2]
3	(a) product of (magnitude of one) force and distance between forces	M1	